$^{12}C(^{28}Si,\alpha p\gamma)$ 2007Ks01,2013Bi10,2020Ra14

2007Ks01: two different experiments: one with E=70 MeV ^{28}Si beam at BARC-TIFR facility and the second with E=88 MeV ^{28}Si beam at NSC facility, both using a target of about 50 μg/cm^2 ^{12}C on a ≈10.5−mg/cm^2 gold backing. γ rays were detected with almost identical (INGA) arrays of eight Compton-suppressed Clover HPGe detectors. Measured Eγ, Iγ, $\gamma\gamma$-coin, $\gamma\gamma(\theta)$(DCO), $\gamma\gamma$(lin pol), Doppler-shift attenuations. Deduced levels, J, π, $T_{1/2}$, γ-ray multipolarities and mixing ratios. Comparisons with available data and large-scale shell-model calculations.

2013Bi10: E=110 MeV ^{28}Si beam was produced from the IUAC-New Delhi accelerator facility. Target was 50 μg/cm^2 thick ^{12}C on 18 mg/cm^2 Au backing. γ rays were detected with the INGA array of thirteen Compton-suppressed Clover detectors. Measured Eγ, Iγ, $\gamma\gamma$-coin, $\gamma\gamma(\theta)$(DCO), Doppler-shift attenuations. Deduced levels, J, π, $T_{1/2}$, evidence for an SD band, γ-ray multipolarities. Comparison with large-scale shell-model calculations, two-level mixing calculation, and with structure of SD states in ^{36}Ar.

2020Ra14: E=110 MeV ^{28}Si beam was produced from the 15UD Pelletron accelerator at the Inter-University Accelerator Centre (IUAC), New Delhi. Target was 50 μg/cm^2 ^{12}C evaporated onto 18 mg/cm^2 Au backing. γ rays were detected with a multi-detector array (Indian National Gamma Array) consisting of 13 Compton-suppressed clovers. Measured Eγ, Iγ, $\gamma\gamma$-coin, $\gamma\gamma$(DCO), $\gamma\gamma$(ADO), $\gamma\gamma$(IPDCO). Deduced levels, J, π, γ-ray multipolarities. Comparisons with theoretical calculations.

Other reactions:

1982Le04: $^{28}Si(^{12}C,X\gamma)$ ^{12}C (E_{cm}=23-35 MeV) beam produced from the University of Washington FN Tandem Van de Graaff. A target of 100 μg/cm^2 ^{28}Si on a thick tantalum backing. A Ge(Li) detector. Measured σ(E), Eγ.

1985Kr21: $^{27}Al(^{12}C,X)$ E=85 MeV/nucleon. Measured particle-particle angular correlations.

^{35}Cl Levels

E(level)[†]	J^π[‡]	$T_{1/2}$[#]	Comments
0.0	3/2$^+$		
1763.0 *6*	5/2$^+$		
2645.8 *6*	7/2$^+$	<0.90 ps	
3003.1 *8*	5/2$^+$		
3163.1 *6*	7/2$^-$		
3942.0 *7*	9/2$^+$	<0.35 ps	
4347.8 *6*	9/2$^-$	0.91 ps *+21−16*	
5407.4 *8*	11/2$^-$	<0.76 ps	
5927.1 *7*	(11/2)$^-$	<0.28 ps	
6087.6 *8*	13/2$^-$		
6119.5? *13*	(11/2$^-$)		
6660.2 *8*	(11/2$^-$)		
7873.3 *8*	13/2$^+$		
8319.4[a] *9*	15/2$^-$	12.5[&] fs *7*	Tentative 8324, (13/2$^-$) level in 2007Ks01 is considered the same level here by the evaluators. $T_{1/2}$: other: <70 fs (2007Ks01). Q(transition)=0.65 *2*, β_2=0.30 (2013Bi10).
8487.8 *10*	15/2$^-$	<0.07 ps	Tentative 8481, (15/2$^-$) level in 2007Ks01 is considered the same level here by the evaluators.
8558.2? *13*	(13/2$^-$)	<0.14 ps	
8787.3? *13*	(15/2$^+$)	<0.28 ps	
8845.1[@] *9*	17/2$^+$		
10181.1[a] *10*	19/2$^-$	42.3[&] fs *28*	Tentative 10181, (19/2$^+$) level in 2007Ks01 is considered the same level here by the evaluators. $T_{1/2}$: other: <0.14 ps (2007Ks01). Q(transition)=0.82 *6*, β_2=0.37 (2013Bi10).
10223.2? *14*	(17/2$^-$)	<0.28 ps	
10862.2? *14*	19/2$^+$	<0.28 ps	
11459.2[@] *12*	21/2$^+$	43.0[&] fs *21*	Q(transition)=0.56 *2*, β_2=0.27 (2013Bi10).
12572.2[a] *12*	23/2$^-$	33[&] fs *4*	Q(transition)=0.71 *5*, β_2=0.32 (2013Bi10).
16306.4[a] *16*	(27/2$^-$)	<9.0[&] fs	Q(transition)>0.5, β_2=0.24 (2013Bi10).

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$^{12}C(^{28}Si,\alpha p\gamma)$ **2007Ks01,2013Bi10,2020Ra14 (continued)**

^{35}Cl Levels (continued)

† From a least-squares fit to γ-ray energies, assuming ΔEγ=1 keV in the fitting procedure.
‡ From the Adopted Levels, unless otherwise noted. Assignments for tentative levels not adopted in Adopted Levels are proposed by 2007Ks01.
# From DSAM in 2007Ks01, unless otherwise noted.
@ Existence of $17/2^+$ and $21/2^+$ levels provides indirect evidence in favor of SD band interpretation and parity-doublet cluster structure (2013Bi10).
& From DSAM in 2013Bi10.
a Band(A): SD band. Proposed configuration=3p−3h, $(sd)^{16}(pf)^3$. Negative-parity partner of SD band in ^{36}Ar (2013Bi10).

$\gamma(^{35}Cl)$

Δ_{IPDCO} corresponds to integrated polarization asymmetry measurement at E=70 MeV at TIFR facility (2007Ks01). The values are from e-mail reply of Dec 1, 2006 from one of the authors (M. Saha Sarkar) of 2007Ks01 to the previous evaluator B. Singh.
Δ_{IPDCO} and DCO under comments are from 2007Ks01, unless otherwise noted. Positive value of Δ_{IPDCO} indicates electric character and negative value indicates magnetic character. Expected value of DCO is ≈1.0 for a stretched transition gated by a transition of same multipolarity, ≈2.0 for dipole when gated by a stretched quadrupole (DCO(Q)) and ≈0.5 for stretched quadrupole when gated by a stretched dipole (DCO(D)) (2007Ks01). Values deviating from above values indicate a mixed transition. DCO values are also available in 2020Ra14, but expected values are opposite to those in 2007Ks01 for stretched dipole and quadrupole transitions when gated by a transition of different multipolarity, that is DCO(Q)≈0.5 for stretched dipole and DCO(D)≈2.0 for stretched quadrupole (2020Ra14).
For ADO ratios under comments from 2020Ra14, expected values are 0.6 for pure dipole transitions and 1.6 for pure quadrupole transitions or ΔJ=0 transitions (2020Ra14).

E_γ†	I_γ‡	E_i(level)	J^π_i	E_f	J^π_f	Mult.#	δ#	Comments
160@	1.1 *3*	3163.1	7/2⁻	3003.1	5/2⁺			I_γ: other: 0.6 *2* at E=70 MeV.
518	9.2 *5*	3163.1	7/2⁻	2645.8	7/2⁺	D		I_γ: other: 10.0 *2* at E=70 MeV. DCO(Q)=0.8 *1*. DCO(D)=1.5 *2*, R_{ADO}=1.6 *4* (2020Ra14).
526		8845.1	17/2⁺	8319.4	15/2⁻			
680	32 *1*	6087.6	13/2⁻	5407.4	11/2⁻	M1(+E2)	+0.1 *1*	I_γ: other: 8.6 *2* at E=70 MeV. DCO(Q)=1.3 *1* at E=70 MeV. Δ_{IPDCO}=−0.09 *2*.
712@&	<1	6119.5?	(11/2⁻)	5407.4	11/2⁻			
779@&	<1	3942.0	9/2⁺	3163.1	7/2⁻			
883	3.2 *2*	2645.8	7/2⁺	1763.0	5/2⁺			Iγ=4.6 *2* at E=70 MeV. Δ_{IPDCO}=+0.01 *5*.
914&	2.4 *5*	8787.3?	(15/2⁺)	7873.3	13/2⁺	D(+Q)	−0.2 *2*	DCO(Q)=2.2 *6* at E=88 MeV.
972	33 *1*	8845.1	17/2⁺	7873.3	13/2⁺	E2		I_γ: other: 2.0 *1* at E=70 MeV. DCO(Q)=1.0 *2* at E=70 MeV, 0.9 *1* at E=88 MeV. Δ_{IPDCO}=+0.08 *4*. DCO(D)=1.8 *2*, DCO(Q)=0.9 *1*, R_{ADO}=1.6 *2* (2020Ra14).
1060	7.3 *7*	5407.4	11/2⁻	4347.8	9/2⁻	M1+E2	+0.2 *1*	I_γ: other: 6.9 *3* at E=70 MeV. DCO(D)=0.9 *1*. Δ_{IPDCO}=−0.10 *4*.
1113		12572.2	23/2⁻	11459.2	21/2⁺			
1185	25 *1*	4347.8	9/2⁻	3163.1	7/2⁻	M1+E2	−0.3 *1*	I_γ: other: 25 *1* at E=70 MeV. DCO(Q)=2.0 *2* at E=70 MeV, 2.8 *3* at E=88 MeV. Δ_{IPDCO}=−0.08 *3*.

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$^{12}C(^{28}Si,\alpha p\gamma)$ 2007Ks01,2013Bi10,2020Ra14 (continued)

$\gamma(^{35}Cl)$ (continued)

E_γ [†]	I_γ [‡]	E_i(level)	J_i^π	E_f	J_f^π	Mult. [#]	δ [#]	Comments
1213		7873.3	13/2+	6660.2	(11/2−)			
1296	0.4 2	3942.0	9/2+	2645.8	7/2+			I_γ: other: 0.2 1 at E=70 MeV.
1336	6 1	10181.1	19/2−	8845.1	17/2+	D(+Q)	−0.2 2	DCO(Q)=2.0 4 at E=88 MeV.
1378 [@&]	2 1	10223.2?	(17/2−)	8845.1	17/2+			
1400	<1	3163.1	7/2−	1763.0	5/2+			
1579	18 2	5927.1	(11/2)−	4347.8	9/2−	(M1+E2)	−0.2 1	I_γ: other: 12 1 at E=70 MeV. DCO(D)=0.9 1 at E=70 MeV, 1.0 1 at E=88 MeV. Δ_{IPDCO}=+0.02 3.
1693		10181.1	19/2−	8487.8	15/2−			
1702	11.4 6	4347.8	9/2−	2645.8	7/2+	E1+M2	+0.5 +5−3	I_γ: other: 10.8 5 at E=70 MeV. DCO(Q)=1.0 2 from 2007Ks01 indicating ΔJ=2 or 0 seems inconsistent with ΔJ=1 from level scheme. Δ_{IPDCO}=+0.04 4. DCO(Q)=0.49 6, R_{ADO}=0.60 6 (2020Ra14).
1740	2.0 2	6087.6	13/2−	4347.8	9/2−	E2		I_γ: other: 2.3 2 at E=70 MeV. Δ_{IPDCO}=+0.12 7.
1763	>9.7	1763.0	5/2+	0.0	3/2+	E2+M1		I_γ: other: >15.7 at E=70 MeV. Δ_{IPDCO}=+0.13 5.
1786	1.8 4	7873.3	13/2+	6087.6	13/2−	D(+Q)	−0.6 6	I_γ: other: 0.1 1 at E=70 MeV. DCO(Q)=1.2 3 at E=88 MeV.
1862		10181.1	19/2−	8319.4	15/2−			
1946	10 1	7873.3	13/2+	5927.1	(11/2)−	E1+M2	+0.2 1	I_γ: other: 0.5 1 at E=70 MeV. DCO(Q)=1.3 2 at E=88 MeV. Δ_{IPDCO}=+0.11 6. DCO(Q)=0.45 8, R_{ADO}=0.59 8 (2020Ra14).
1985 [@]	2.0 4	5927.1	(11/2)−	3942.0	9/2+			I_γ: other: 1.6 6 at E=70 MeV.
2017 [@&]	7 1	10862.2?	19/2+	8845.1	17/2+			
2179	6.5 7	3942.0	9/2+	1763.0	5/2+	E2		I_γ: other: 11.1 6 at E=70 MeV. Δ_{IPDCO}=+0.07 5.
2232		8319.4	15/2−	6087.6	13/2−			E_γ: from 2013Bi10 only.
2244	61 2	5407.4	11/2−	3163.1	7/2−	E2		I_γ: other: 50 3 at E=70 MeV. DCO(Q)=1.0 1 at E=70 MeV, 1.2 1 at E=88 MeV. Δ_{IPDCO}=+0.04 2.
2312		6660.2	(11/2−)	4347.8	9/2−			
2391		12572.2	23/2−	10181.1	19/2−			
2400	<1	8487.8	15/2−	6087.6	13/2−	D(+Q)	+0.2 +3−2	E_γ: other: 2394 (2007Ks01). DCO(Q)=1.4 4 at E=88 MeV.
2466	22 2	7873.3	13/2+	5407.4	11/2−	E1+M2	−0.25 10	I_γ: other: 1.2 1 at E=70 MeV. DCO(Q)=2.4 3. Δ_{IPDCO}=+0.21 6. DCO(Q)=0.51 8, R_{ADO}=0.65 9 (2020Ra14).
2614		11459.2	21/2+	8845.1	17/2+			
2631 [@&]	<1	8558.2?	(13/2−)	5927.1	(11/2)−	D(+Q)	−0.2 +2−3	DCO(Q)=2.4 7 at E=88 MeV.
2646	37 2	2645.8	7/2+	0.0	3/2+	E2		I_γ: other: 47 2 at E=70 MeV. Δ_{IPDCO}=+0.08 5. DCO(D)=1.7 2, R_{ADO}=1.6 2 (2020Ra14).
2757		8845.1	17/2+	6087.6	13/2−			
2764		5927.1	(11/2)−	3163.1	7/2−			E_γ: from 2013Bi10 only.

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$^{12}C(^{28}Si,\alpha p\gamma)$ 2007Ks01,2013Bi10,2020Ra14 (continued)

$\gamma(^{35}Cl)$ (continued)

E_γ [†]	I_γ [‡]	E_i(level)	J_i^π	E_f	J_f^π	Mult. [#]	δ [#]	Comments
2912	<1	8319.4	15/2⁻	5407.4	11/2⁻			E_γ: other: 2917 (2007Ks01, tentative). Mult.,δ: D(+Q) with δ=−0.1 +3−2 if J(8319)=13/2 in 2007Ks01. DCO(Q)=1.9 6 at E=88 MeV.
3003 [@]	>1.1	3003.1	5/2⁺	0.0	3/2⁺			I_γ: other: >0.6 at E=70 MeV.
3080	<1	8487.8	15/2⁻	5407.4	11/2⁻			E_γ: other: 3074 (2007Ks01).
3163	100 5	3163.1	7/2⁻	0.0	3/2⁺	M2+E3	+0.16 1	I_γ: other: 100 2 at E=70 MeV. DCO(Q)=0.9 1 at E=70 MeV, 0.7 1 at 88 MeV. Δ_{IPDCO}=−0.06 2.
3497		6660.2	(11/2⁻)	3163.1	7/2⁻			
3734		16306.4	(27/2⁻)	12572.2	23/2⁻			
4347 [@]	<1	4347.8	9/2⁻	0.0	3/2⁺			I_γ: other: <1 at E=70 MeV.

[†] From 2013Bi10, unless otherwise noted.

[‡] From 2007Ks01, measured at E(^{28}Si)=88 MeV at the NSC. Values are also available from E=70 MeV data of 2007Ks01 at TIFR facility, but less complete and are given under comments where available.

[#] Q and D+Q with δ(Q/D) are from $\gamma\gamma(\theta)$(DCO) in 2007Ks01 and magnetic/electric characters are from $\gamma\gamma$(lin pol) in 2007Ks01 where available and/or RUL where level $T_{1/2}$ is measured, unless otherwise noted.

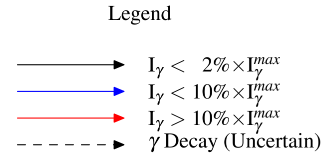
[@] From 2007Ks01 only.

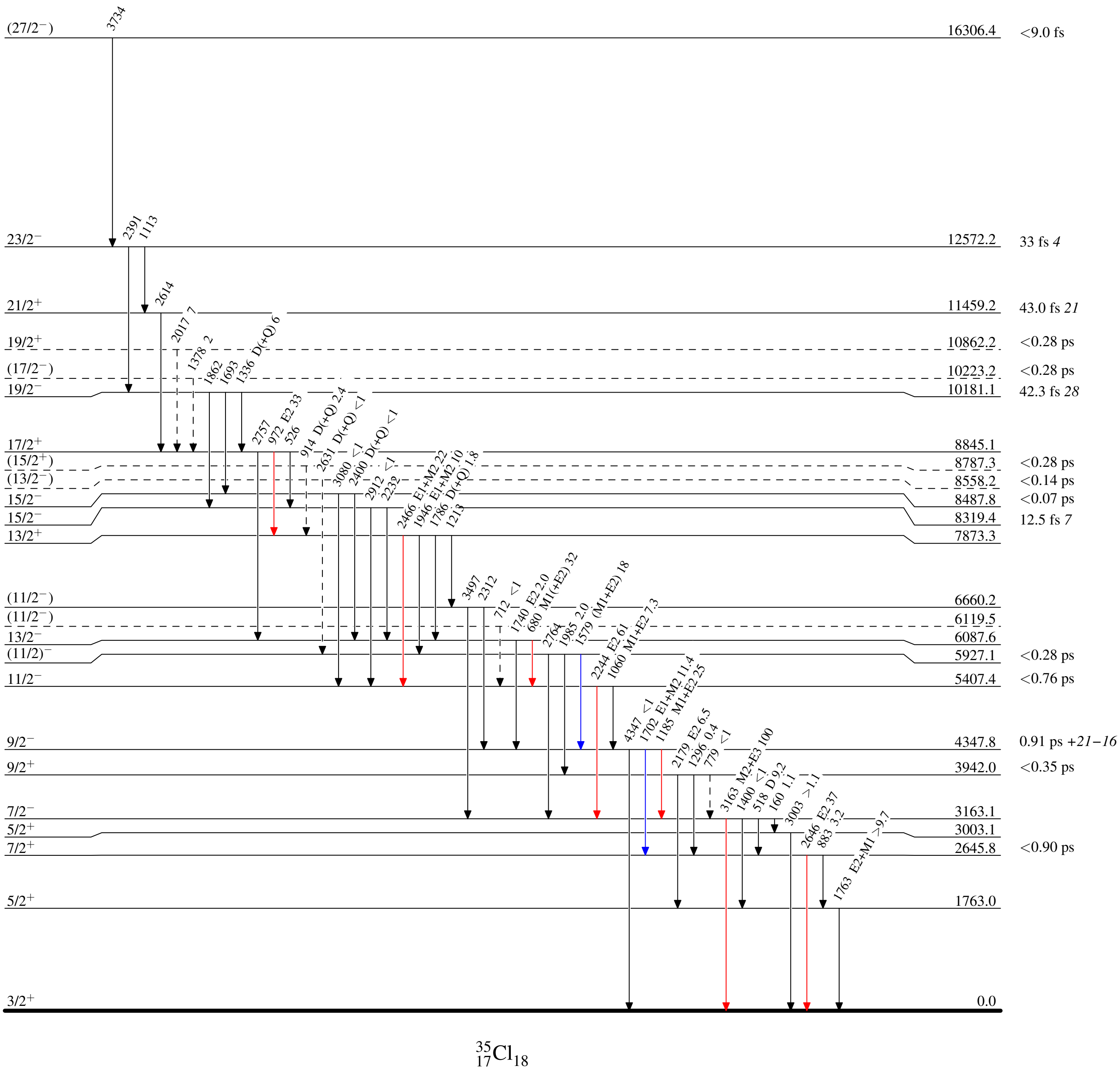
[&] Placement of transition in the level scheme is uncertain.

^{12}C(^{28}Si,αpγ) 2007Ks01,2013Bi10,2020Ra14

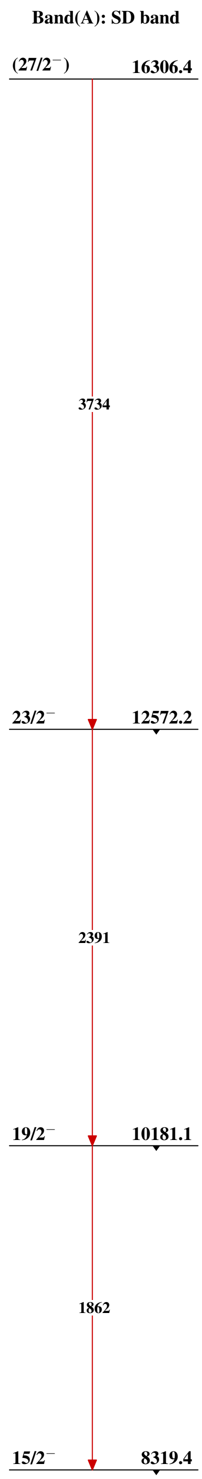
Level Scheme

Intensities: Relative I_γ





$^{12}C(^{28}Si,\alpha p\gamma)$ 2007Ks01,2013Bi10,2020Ra14



$^{35}_{17}Cl_{18}$